Multimodal Sentiment and Emotion Analysis

Multi-Layer Feature Fusion and Multi-Task Learning

Manas Inamdar Soham Jahagirdar Parth Tokekar Gautam Gandhi

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Team TokeNization
Introduction to NLP (S26) — Final Submission

Outline

1. Motivation & Problem
2. Related Work & Datasets
3. Paper Proposal
4. Architecture Overview
5. UFEN: Design & Rationale
6. Cross-Modal Attention
7. MTFN & Encoder-Decoder
8. Multi-Task Loss
9. Implementation Decisions
10. Phase 1: CMU-MOSI Results
11. Phase 2: MELD Experiments
12. Best Result (Exp 9.1)
13. Analysis & Ablations
14. Limitations & Future Work
15. Conclusion

Paper re-implemented: Cai et al., *Sci. Reports* 2025 (UFEN+MTFN).

Motivation

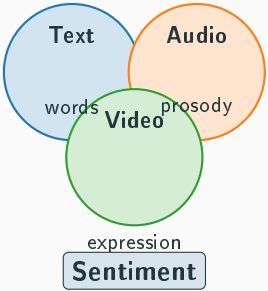
Why Multimodal?

Human communication is multimodal:

- **Text** — what is said
- **Audio** — how it is said
- **Video** — facial reaction

"It was fine."

Text only	→ neutral
+ flat voice	→ negative
+ frown	→ disappointed



Three Core Challenges

1. Heterogeneity

Different feature spaces: BERT 768-d, COVAREP 74-d, FACET 47-d

2. Conflict

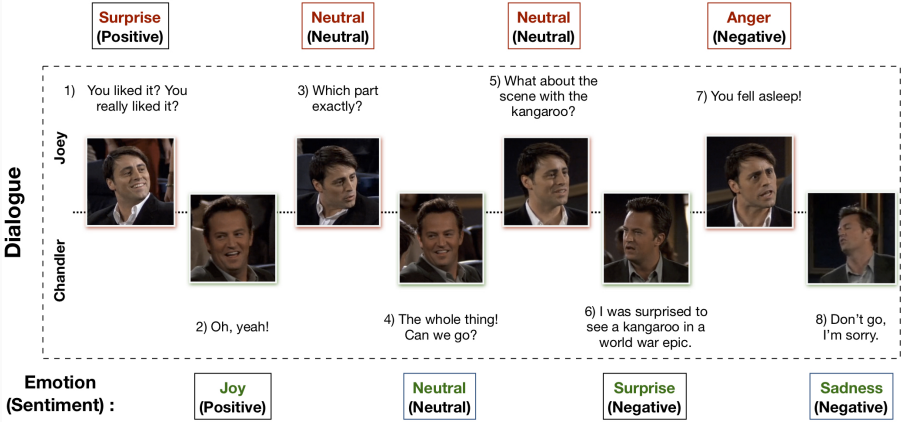
Modalities can disagree (sarcasm, irony); simple averaging cancels signal

3. Text Dominance

Text alone $\approx$ 73% on MOSI; model tends to ignore A/V

UFEN+MTFN addresses all three with a two-stage pipeline and five-task supervision.

Emotion Shifts in Dialogue



Emotions shift rapidly within a conversation — each utterance can carry a different emotion and sentiment. The model must handle **intra-speaker** and **inter-speaker** shifts using all three modalities.

Example from MELD (Friends): Joey and Chandler's emotions shift across 8 turns.

Related Work & Datasets

Related Work

Category	Representative Methods
Per-modality encoders	BERT, LSTM/GRU, COVAREP, FACET
Tensor fusion	TFN, LMF
Attention fusion	MulT, MFN, MISA
Multi-task MSA	Self-MM, MMIM
ERC on MELD	DialogueRNN, DialogueGCN, MM-DFN, MultiEMO
Our base paper	**UFEN+MTFN** (Cai et al., 2025)

UFEN+MTFN combines MulT-style cross-modal attention with Self-MM-style multi-task learning and adds a Transformer encoder–decoder on top.

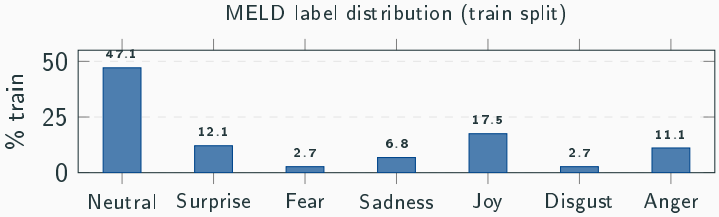
Datasets

CMU-MOSI (Phase 1)

- 2,199 YouTube review clips
- Regression: $[-3, +3]$
- Splits: 1,283 / 214 / 686
- Text: BERT-768 (fine-tuned)
- Audio: COVAREP 74-d
- Visual: FACET 47-d

MELD (Phase 2)

- 13,708 *Friends* utterances
- 7-class classification
- Splits: 9,988 / 1,108 / 2,610
- Strongly imbalanced



Architecture

What the Paper Proposes: Cai et al. (2025)

Three problems with existing methods:

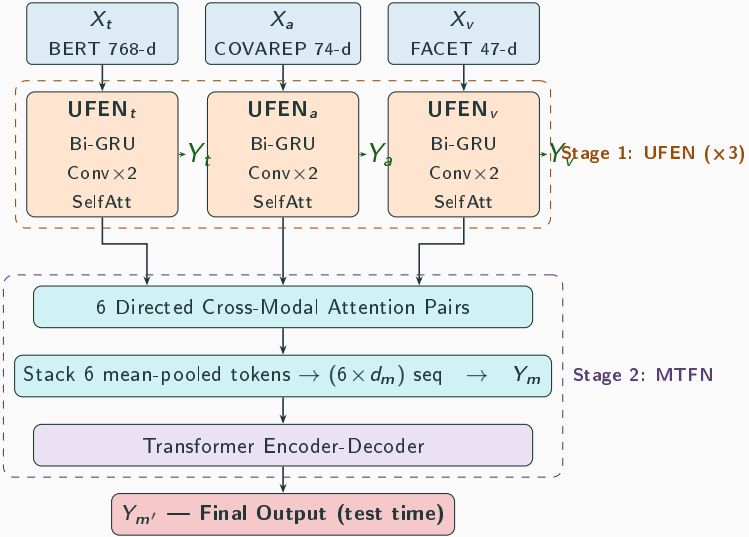
1. **Shallow unimodal features** — a single GRU or LSTM misses multi-scale patterns within one modality
2. **One modality dominates** — text takes over; audio and video get ignored during fusion
3. **Inconsistent cross-modal sentiment** — face says *negative*, words say *positive*; model struggles to reconcile

Two architectural solutions:

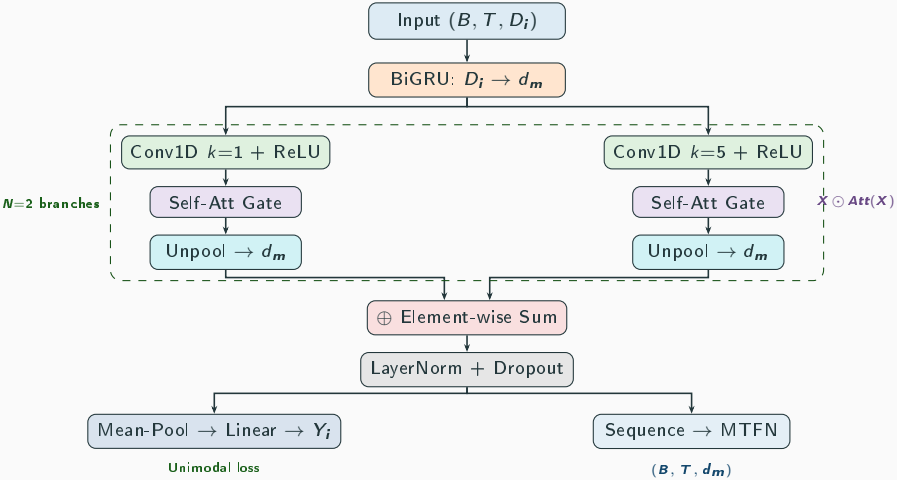
- **UFEN** (one per modality):
  BiGRU + parallel Conv1D + self-attention gating → richer, multi-scale per-modality representation
- **MTFN**:
  All-pairs cross-modal attention + Transformer enc-dec + 5-head multi-task training → balanced, complementary fusion

Evaluated on CMU-MOSI, CMU-MOSEI, CH-SIMS. MOSI paper target: MAE 0.728, Acc-2 np 86.6%, F1 86.7%.

Architecture Overview



Stage 1: UFEN — Architecture



Our values: d_m=128, kernels [1, 5], conv_dim= 64. Paper specifies N=2 but not the kernel sizes.

Stage 1: UFEN — Intuition

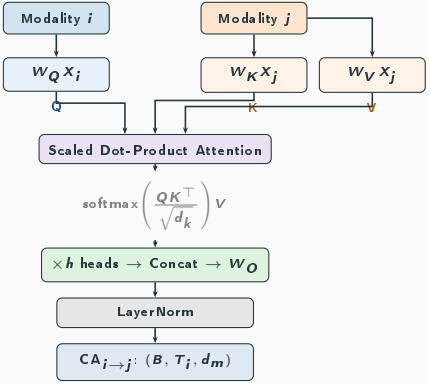
Component	What it does and why we need it
Bi-GRU	Reads the sequence forward *and* backward — a word's meaning depends on both past and future context. Maps raw features (D_i) to a shared hidden dimension (d_m).
Conv1D k=1	Pointwise filter — captures per-token features with no temporal window.
Conv1D k=5	5-step window — captures short phrases and local temporal patterns (e.g. "not really happy").
Self-Att Gate	$X \odot \mathrm{Att}(X)$ — **multiplicative**: attention scores *suppress* unimportant positions rather than adding to them. Irrelevant timesteps get zeroed out.
Unpool + Sum	Each branch is projected back to d_m, then summed. Merges multi-scale views while keeping dimension fixed.
Y_i head	Mean-pool $\rightarrow$ Linear: a unimodal prediction head. Forces UFEN to carry enough sentiment signal *independently* of fusion. Without this loss, the model ignores audio and video.

Cross-Modal Attention: Modalities Learn from Each Other

Intuition: modality i asks modality j —
"What in you is relevant to my positions?"

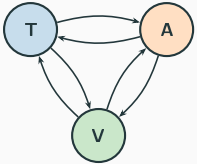
- Modality $i \to$ **Query**
- Modality $j \to$ **Key + Value**
- Output: i's representation enriched by j

Example — $t \to a$:
The word "fine" attends to flat-pitch audio $\to$ model learns "fine" said monotonously means *negative*.
$\mathrm{CA}_{i\to j} = \mathrm{LN}(\mathrm{MhAtt}(Q_i, K_j, V_j))$
6 ordered pairs; output (B, T_i, d_m) each.
No residual connection in our implementation.



Stage 2: MTFN — Fusion and Reconstruction

Step 1: 6 cross-attention pairs



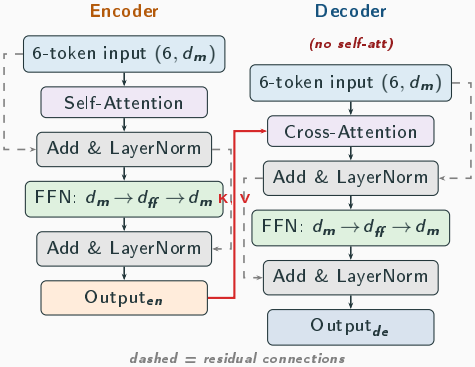
Step 2: Mean-pool each CA → 6 vectors of d_m. Stack as sequence.

Step 3: Y_m — Flatten $6d_m$ → Linear.

Step 4: **Encoder** — Self-attention over the 6 tokens.

Step 5: **Decoder** — Cross-attends to encoder. **No self-attention** (Eq. 13).

Step 6: $Y_{m'}$ — Mean-pool decoder → Linear → **final prediction**.



Multi-Task Loss: 5 Heads, 1 Objective

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_t + \mathcal{L}_a + \mathcal{L}_v}_{\text{unimodal}} + \underbrace{\mathcal{L}_m + \mathcal{L}_{m'}}_{\text{multimodal}}$$

Why 5 losses?

- Without unimodal losses, model ignores A/V
- Text alone reaches ∼73% on MOSI
- Forces each branch to carry signal independently

Task-specific variants

- Phase 1 (MOSI): MSE loss, 5 regression heads
- Phase 2 (MELD): Weighted cross-entropy + label smoothing, 5 classification heads

Test-time output is $Y_{m'}$ only; the other four heads exist only during training.

Our Implementation: Key Decisions Where the Paper Was Silent

Aspect	Paper says	Our choice and reason
Prediction head	"softmax($Wx+b$)" for all heads	**Plain Linear** — this is regression with MSE; softmax would constrain output to $(0, 1)$
Decoder structure	Eq. 13: only cross-attention	**No self-attention in decoder** — we follow the equation literally; standard decoders have it but the paper doesn't
Conv kernel sizes	N=2 branches; no sizes given	$k = [1, 5]$ — pointwise + 5-gram context; tuned up from initial $[1, 3]$
Hidden dim d_m	Not stated	**128** — 4 cross-att heads $\times$ 32-d each; scales to the small dataset
Mean-pool step	Not explicit	**Required** — CA outputs have different sequence lengths T; only way to form a fixed 6-token sequence
BERT learning rate	Single lr $= 5\times10^{-3}$ for all params	**lr_bert** $= 2\times10^{-5}$ — standard fine-tuning; 5e-3 destroys pretrained BERT weights

Phase 1: CMU-MOSI

Phase 1: CMU-MOSI — Best Configuration (P3-2)

Search scope: 25+ experiments in 3 sub-phases

- **1a**: Single-param sweeps
- **1b**: Combinations
- **1c**: Regularisation

Key unlocks

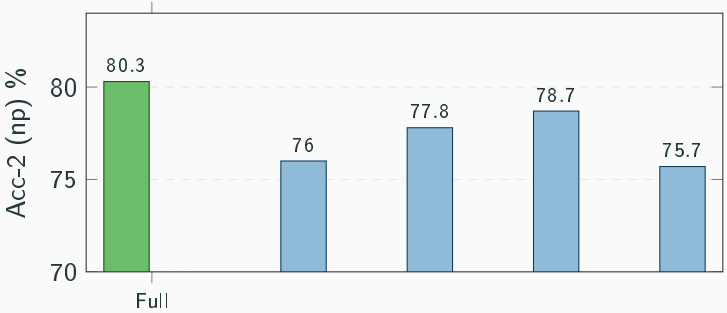
- $d_{\text{ff}} = 128$ (not $2 \times d_m$)
- Kernels $[1, 5]$
- Cosine LR + early stop = 15
- BERT warmup (5 epochs linear)
- Weight decay was catastrophic

Metric	Ours (P3-2)	Paper
MAE ↓	**0.812**	0.728
Corr ↑	**0.745**	0.792
Acc-7 ↑	**43.0**	46.7
Acc-2 nn ↑	**78.6**	85.2
Acc-2 np ↑	**80.3**	86.6
F1 np ↑	**80.4**	86.7

Dev MAE reached 0.7434 — within 0.015 of paper's target.

∼110.8M params (BERT dominates).

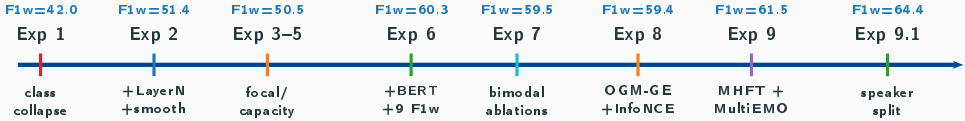
Phase 1 Ablation — Every Component Matters



- **Cross-modal attention** −4.3 points → biggest component
- **Unimodal losses** −2.5 points → stop text collapse
- **Encoder–decoder** −1.6 points → refinement helps
- **Single conv** −4.6 points → multi-scale is essential

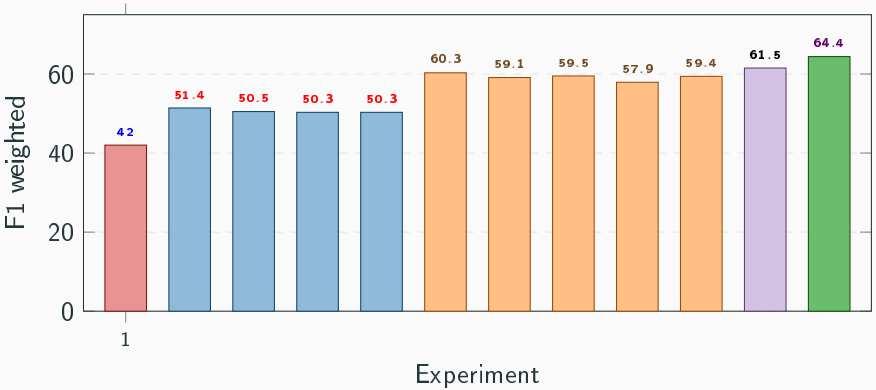
Phase 2: MELD

Phase 2 Experiment Trail — Timeline



Three major jumps: Exp 2 (feature norm), Exp 6 (BERT), Exp 9.1 (MHFT + clean split).

Phase 2 Results — Weighted F1 Progression



Exp 1: collapse **2–5**: GloVe plateau **6–8**: BERT (111M params) **9**: MHFT **9.1**: **best**

Exp 1–2: Diagnosing Class Collapse

Exp 1 failure: only Neutral & Anger predicted

- LR 5e-3 too high (oscillation)
- GloVe L2 norm ≈ 5 vs. audio 0.85, video 0.72 $\rightarrow$ **7× text dominance**
- Trivial minimum: majority class only

Exp 2 fix: 3 targeted changes

- Per-modality LayerNorm
- LR $\rightarrow 10^{-3}$
- Label smoothing 0.1

Result

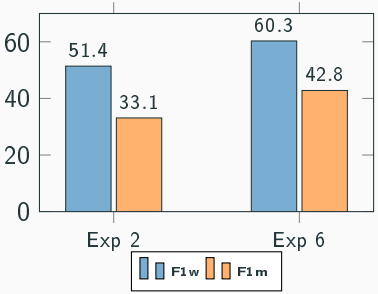
- F1w: 42.0 $\rightarrow$ 51.4 (**+9.4**)
- F1m: 19.6 $\rightarrow$ 33.1 (**+13.5**)
- All 7 classes get predictions

Exp 6: BERT Breaks the GloVe Ceiling

Why GloVe plateaued: context-free embeddings cannot disambiguate

- *"Yeah, right"* — sarcasm or agreement?
- *"I can't believe it"* — joy, anger, surprise?

Exp 6: `bert-base-uncased`, fine-tuned end-to-end, differentiated LR (BERT 2e-5, rest 1e-3), warmup + cosine.

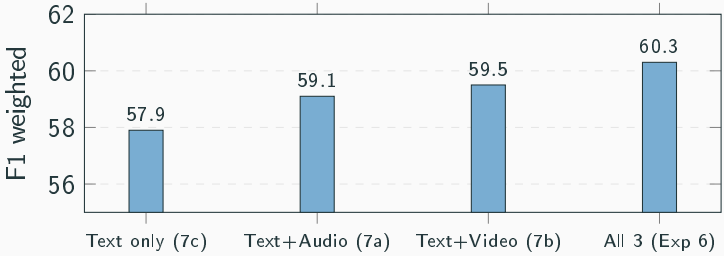


+8.9 F1w,

+9.7 F1m

(every class improved)

Exp 7: Bimodal Ablations — Who Contributes What?



- **Text alone** = 57.9 F1w — very close to full 3-modal (60.3)
- Audio & video together add only **+2.4 F1w** over text
- Audio shines on **Sadness** (prosody); video on **Neutral** (face grounding)
- *Friends* is scripted — emotion is encoded in word choice

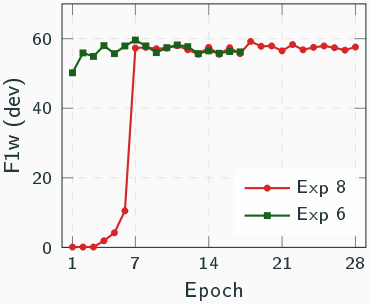
Exp 8: OGM-GE + InfoNCE + Curriculum — A Wash

Three research-backed interventions stacked:

1. **OGM-GE** (Peng et al., CVPR 2022): rescale weak-branch gradients
2. **InfoNCE** (CLIP-style): contrastive alignment across modality pairs
3. **Curriculum**: freeze text → unfreeze → full

Result: F1w 60.3 → 59.4 (−0.9)

Diagnosis: Phase A (A+V only) gave **2% accuracy** over 3 epochs — the weak modalities cannot bootstrap alone on MELD.



Phases A & B are wasted compute.

Exp 9: MHFT Front-End — The Breakthrough

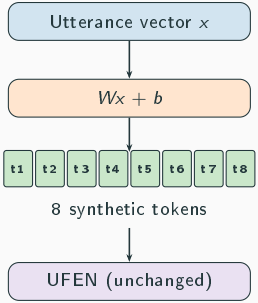
Problem: MultiEMO features are utterance-level (one vector per clip) — UFEN expects a time sequence.

Solution: **MHFT**. Project 1 vector $\rightarrow K{=}8$ tokens:

$$\mathrm{MHFT}(x) = \mathrm{reshape}(Wx + b) + P \in \mathbb{R}^{8 \times d_m}$$

UFEN and MTFN run **unchanged** on the synthetic sequence.

Features: RoBERTa-768 (T) + openSMILE-512 (A) + DenseNet-1000 (V)



F1w: 60.3 $\rightarrow$ 61.5

Params: 111M $\rightarrow$ 3.58M (30$\times$ shrink)

Exp 9.1: Closing the Dev→Test Gap

Problem: Exp 9 had dev F1w = 71.7 but test = 61.5 — a **10-point** generalisation gap.

Root cause: Random split of `trainVid` leaked *Friends* speakers into dev. Model memorised speaker-specific patterns.

Three fixes:

1. Speaker-disjoint split

Hold out whole dialogue groups by dominant speaker

Gap: 10.2 → 5.7

2. Capped class weights

Cap Fear/Disgust weights at 1.5 (was 2.24 / 2.41)

False positives: 84 → 43

3. Top-5 ensemble

Average softmax of top-5 dev checkpoints

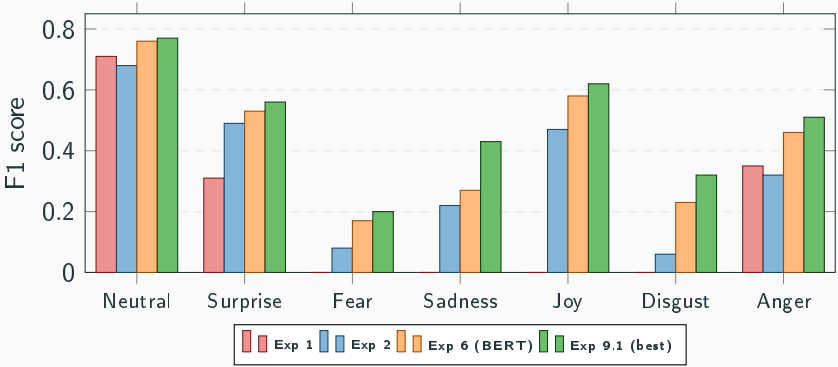
+0.9 F1m free

	Exp 9	**Exp 9.1**	**Δ**
Test F1w	61.5	**64.4**	+2.9
Test F1m	45.7	**49.0**	+3.3
Train size	9,920	6,464	−3,456

Less training data, better generalisation.

Analysis

Per-Class F1 — Exp 9.1 vs. Baselines



Exp 9.1 wins or ties on **5/7 classes**. Largest gains: Sadness (+0.08), Disgust (+0.07), Anger (+0.04).

What Did *Not* Help — and Why

Attempted	Δ F1w	Root cause
Focal loss ($\gamma = 2$)	-0.9	Over-penalises easy Neutral/Joy, hurts mid-frequency classes
Video projection 2048$\rightarrow$256	-1.1	BiGRU implicitly selects channels end-to-end; fixed projection discards
Doubled capacity ($d_m = 256$)	-1.1	7.1M params on 10K samples $\rightarrow$ instability (best ep = 3)
Lower LR (5×10^{-4})	-1.1	Slower convergence, no better final
Scheduled unfreezing	-0.9	A/V cannot bootstrap alone (2% accuracy in Phase A)

Lesson: on small datasets with one dominant modality, small surgical fixes beat elaborate interventions.

Error Analysis — Exp 9.1 Confusion Patterns

Three persistent patterns:

1. **Neutral absorbs errors** — ∼7–10% of every other class pulled into Neutral
2. **Anger/Surprise/Joy triangle** — shared high-arousal prosody
3. **Fear & Disgust** still hard (50 & 68 test samples)

	Neu	Sur	Fear	Sad	Joy	Dis	Ang
Neutral	**912**	65	43	60	84	28	64
Surprise	24	**177**	4	4	27	10	35
Fear	9	4	**17**	6	5	1	8
Sadness	40	14	26	**83**	12	6	27
Joy	59	40	8	6	**253**	10	26
Disgust	14	5	4	2	0	**28**	15
Anger	44	41	17	13	29	22	**179**

Diagonal = correct predictions; green = strong, orange = weak.

Wrapping Up

Key Takeaways

1. **UFEN+MTFN architecture is sound** — every component is ablation-justified on MOSI.
2. **BERT is the dominant lever on MELD**: context-free embeddings plateau at ∼51 F1w; fine-tuned BERT jumps it to 60.3.
3. **MHFT is the surprise winner**: 3.58M params beats 111M BERT on every MELD metric. Pre-extracted features + a lightweight tokenization bridge is a very favourable trade.
4. **Evaluation integrity matters as much as architecture**: swapping a random dev split for speaker-disjoint closed a 4-point gap with *less* training data.
5. **Negative results are informative**: focal loss, doubled capacity, and curriculum unfreezing all failed — each teaches something about the dataset.

Limitations & Future Work

Limitations

- Small datasets (MOSI: 1.3K; MELD: 10K train)
- Single-seed reporting (no std)
- Paper hyperparameters under-specified
- Wave2Vec2-32 & ResNet-101 are not SOTA for affect
- No inter-utterance context

Future Work

- Multi-seed mean $\pm$ std
- Stronger A/V extractors: WavLM, Whisper, AffectNet
- Dialogue context (DialogueRNN-style)
- MHFT on time-aligned features
- Extend to MOSEI, IEMOCAP, CH-SIMS

Ethical note: MELD is drawn from *Friends*, a US sitcom with a narrow demographic and scripted emotion. Deployed systems need re-validation on diverse populations and spontaneous speech.

Thank you!

Questions?

Code: `github.com/Gautam-Gandhi/multimodal-sentiment-analysis`

Team TokeNization: **Manas**, **Soham**, **Parth**, **Gautam**